

Variadic Friends

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1 Introduction

This paper proposes support for granting friendship to all classes in a parameter pack. Several existing idioms are implemented by providing friendship to a class via template parameter. However, these patterns can only be used with a single template parameter, because friendship cannot be currently granted to a pack of types.

The following code snippet provides a general summary of the current issue.

```
template <typename T>
class Foo1
{
    friend T;           // Supported
public:
    // ...
};

template <typename... Ts>
class Foo2
{
    friend Ts...;       // ERROR
public:
    // ...
};
```

2 Motivation

This section documents two common idioms, and how they could be expanded with support for pack expansion in a friend declaration.

2.1 The Passkey Idiom

Granting general friendship has wide implications. In the following example, `Blarg` has been granted friendship by `Foo`.

```
class Blarg;
class Foo
{
    friend Blarg;
    // ...
};
```

Thus, `Blarg` can access all private members of `Foo`, even if friendship were only needed in order to access a single non-public member. Wide friendship has a number of issues, and is not generally desired.

With the Passkey idiom, limited access can be granted on a per-member function basis. In the following example, the `Foo::do_something` member function is public, but it must be given an instance of `Passkey` as a function argument.

```
class Foo
{
public:
    // Only callable from Blarg
    void do_something(Passkey);
};
```

If `Passkey` has a non-public constructor, but grants friendship to `Blarg`, then `Blarg` can call `Foo::do_something` by passing an instance of `Passkey`.

```
class Passkey
{
    friend Blarg;
    Passkey() = default;
};
```

Furthermore, the only entities that can call `Foo::do_something` are those who have been granted friendship with `Passkey`, since friendship with `Passkey` is required in order to create an instance of `Passkey`.

This technique can be generalized with the following idiom.

```
template <typename T>
class Passkey
{
    friend T;
    Passkey() = default;
};

class Foo
{
public:
    // Only callable from Blarg
```

```
void do_something(Passkey<Blarg>);
};
```

The above code has the same effect, where `Foo::do_something` can only be called if the caller can create an instance of `Passkey<Blarg>`.

We would like to expand this idiom, and grant access to `Foo::do_something` by more than one class.

```
template <typename... Ts>
class Passkey
{
    friend Ts...;
    Passkey() = default;
};

class Foo
{
public:
    // Only callable from Blarg, Blip, and Baz
    void do_something(Passkey<Blarg, Blip, Baz>);
};
```

However, C++ does not allow the variadic friend declarations, so the above code will not compile.

2.2 CRTP Access to Derived

Another common pattern is to inherit from some class template, passing the type of the derived class as a template parameter to the base class.

There may be parts of the derived class API which are needed in the base class, but only the base class, so they are private, and friendship is granted to the base class.

```
template <typename DispatcherT, typename MsgT>
class Receiver
{
    // Can static_cast to DispatcherT and access private stuff
};

template <typename MsgT>
class Dispatcher
: public Receiver<Dispatcher<MsgT>, MsgT>
{
    friend Receiver<Dispatcher, MsgT>;

    // stuff that we want Receiver to access
    // ...

public:
    using Receiver<Dispatcher, MsgT>::Receiver;
};
```

If we want to support multiple base classes, we would like to write the following.

```
template <typename DispatcherT, typename MsgT>
class Receiver
{
};

template <typename... MsgTs>
class Dispatcher
: public Receiver<Dispatcher<MsgTs...>, MsgTs>...    // OK
{
    friend Receiver<Dispatcher, MsgTs>...;           // ERROR

    // stuff that we want any Receiver to access
    // ...

public:
    using Receiver<Dispatcher, MsgTs>::Receiver...; // OK
};
```

Unfortunately, this results in a compiler error, because pack expansion is not supported for friend declarations. Note, however, that both inheritance and `using` support pack expansion.

3 Other Implications

The following would be implicitly supported as a result of supporting pack expansion for friend declarations.

```
class Phoebe
{
    friend Rachel, Monica, Ross, Chandler, Joey;
};

template <typename T, typename U>
class Foo
{
    friend T, U;
};

template <typename... Ts, typename... Us>
class Bar<mp_list<Ts...>, mp_list<Us...>>
{
    friend Ts..., Us...;
};
```

4 Wording

These changes are relative to C++ draft n4944.

Extend the grammar for a `friend` declaration in 11.8.4 [class.friend]/3 and extend the example:

- ³ A friend declaration that does not declare a function shall have one of the following forms:

```

+ friend-typename-specifier-list:
+   typename-specifier ...opt
+   friend-typename-specifier-list, typename-specifier ...opt

friend elaborated-type-specifier ;
friend simple-type-specifier ;
- friend typename-specifier ;
+ friend friend-typename-specifier-list ;

```

And extend the example following that paragraph:

```

class C;
typedef C Ct;

class X1 {
    friend C;                // OK, class C is a friend
};

class X2 {
    friend Ct;               // OK, class Ct is a friend
    friend D;                // error: D not found
    friend class D;          // OK, elaborated-type-specifier declares new class
};

template <typename T> class R {
    friend T;
};

+ template <typename... Ts> class V {
+   friend Ts...;
+ };

R<C> rc;                     // class C is a friend of R<C>
R<int> Ri;                   // OK, "friend int;" is ignored
+ V<C, X1> vc;               // both C and X1 are friends of V<C, X1>

```

Insert a new stipulation in 13.7.4 [temp.variadic]/5:

- ⁵ A pack expansion consists of a pattern and an ellipsis, the instantiation of which produces zero or more instantiations of the pattern in a list (described below). The form of the pattern depends on the context in which the expansion occurs. Pack expansions can occur in the following contexts:
- (5.13) — In a friend-declaration ([class.friend]); the pattern is a `_typename-specifier_`.

5 Acknowledgements

I had been sitting on this since posting to the mailing list about it in January 2020. This paper was finally written during C++Now 2023, Feature in a Week. Thus, it is very appropriate to say that it would not exist without that program. Thanks to Jeff Garland, Marshall Clow, Barry Revzin, and JF Bastien for running that program, and all the attendees who helped discuss and review the proposal. Special thanks to Barry Revzin for lots of help with the document itself.